

## Quantum Field Theory – Homework 2

Name: \_\_\_\_\_ Due Date: 2025.11.03

---

*The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.*

### Problem 1. Probability Interpretation and the Klein–Gordon Equation (20 points)

In non-relativistic quantum mechanics, the Schrödinger equation

$$i \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{1}{2m} \nabla^2 \psi(\vec{x}, t)$$

admits a conserved quantity interpreted as probability. Multiplying by  $\psi^*$  and subtracting the complex conjugate gives

$$\frac{\partial}{\partial t} |\psi|^2 + \nabla \cdot \left[ \frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi) \right] = 0.$$

Hence we can identify

$$\rho = |\psi|^2, \quad \vec{J} = \frac{i}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi),$$

which satisfy the continuity equation  $\partial_t \rho + \nabla \cdot \vec{J} = 0$ , so that  $\rho$  is real, non-negative, and represents the probability density.

Now consider the relativistic Klein–Gordon equation for a single relativistic particle

$$\partial_\mu \partial^\mu \psi(\vec{x}, t) - m^2 \psi(\vec{x}, t) = 0.$$

- (a) Derive the conserved current associated with this equation by combining it with its complex conjugate. Show that it can be written as

$$\partial_\mu j^\mu = 0, \quad j^\mu = i(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*).$$

Identify the time component  $j^0$  and spatial component  $\vec{j}$ .

- (b) Examine the expression for  $j^0$  and discuss its sign. Why can  $j^0$  take negative values even for well-behaved wave packets? Explain why it therefore cannot be interpreted as a probability density.

- (c) Briefly explain the physical implication of this result. Why does the failure of a positive-definite probability density indicate that the Klein-Gordon equation cannot consistently describe a single relativistic particle, and motivates the need for a field-theoretic description?

**Problem 2. Noethers Theorem and Energy-Momentum Conservation (20 points)**

Consider the real scalar field theory

$$S[\phi] = \int d^4x \left[ -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right].$$

- (a) Show that the action is invariant under spacetime translations  $x^\mu \rightarrow x^\mu + a^\mu$ .
- (b) Using Noethers theorem, derive the conserved current associated with this symmetry and identify the corresponding conserved quantity.
- (c) Write down the components of the energy-momentum tensor  $T^{\mu\nu}$  for this theory.

**Problem 3. Lorentz Invariance of the Scalar Field Action (15 points)**

Show explicitly that the scalar field action

$$S = \int d^4x \left[ -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right]$$

is invariant under an infinitesimal Lorentz transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \quad \phi(x) \rightarrow \phi'(x) = \phi(x),$$

where  $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$ . Then, use Noether's theorem to find the expression for the conserved angular-momentum current  $M^{\lambda\mu\nu}$ .

**Problem 4. From Discrete Systems to Field Theory (15 points)**

Starting from the Lagrangian of a one-dimensional chain of coupled oscillators

$$L = \sum_n \left[ \frac{1}{2} \mu \dot{\eta}_n^2 - \frac{\lambda}{2} (\eta_{n+1} - \eta_n)^2 - \frac{1}{2} \sigma \eta_n^2 \right],$$

derive its continuum limit as the lattice spacing  $a \rightarrow 0$ , obtaining a field theory in terms of a continuous field  $\eta(x, t)$ . Identify the effective mass  $m$  and propagation speed  $v$ , and discuss under what condition this theory becomes relativistic.